

D 112341

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Name.....

Reg. No.....

**FIRST SEMESTER (CUFYUGP) DEGREE EXAMINATION
NOVEMBER 2024**

Mathematics

MAT 1CJ 101—DIFFERENTIAL CALCULUS

(2024 Admission onwards)

Time : Two Hours

Maximum Marks : 70

Section A*All questions can be answered.**Each question carries 3 marks.**(Ceiling 24 marks)*

1. Draw the graph of the equation $y^2 = 4x$.
2. Write any three properties of absolute value function $f(x) = |x|$.
3. Find $\lim_{x \rightarrow +\infty} \left(\frac{1 - e^x}{1 + e^x} \right)$.
4. Does $\lim_{x \rightarrow 1} \frac{1}{1-x} = \lim_{x \rightarrow 0^-} \frac{1}{x}$ exist ? Explain.
5. At what points are the function $\frac{x}{\sin x}$ continuous ?
6. Let $f(x) = x^2$ and $g(x) = x + 6$. Find $(g \circ f)(x)$ and $(g \circ f)'(x)$.
7. Find the absolute maximum and minimum values of $f(x) = 4x^2 - 12x + 10$ on the closed interval $[1, 2]$.
8. Find the two x -intercepts of the function $f(x) = x^2 - 5x + 4$ and confirm that $f'(c) = 0$ at some point c between those intercepts.
9. Locate the critical points of the function $f(x) = x^3 - 3x^2$ and identify which critical points are stationary points.
10. Determine whether the statement "If the graph of f has a cusp at $x = 1$, then f cannot have an inflection point at $x = 1$ " is true or false. Explain your answer.

Turn over

Section B

*All questions can be answered.
Each question carries 6 marks.
(Ceiling 36 marks)*

11. Find the center and radius of the circle $x^2 + y^2 + 4x - 6y - 3 = 0$.
12. Determine whether the statement “If $\lim_{x \rightarrow a} f(x)$ and $\lim_{x \rightarrow a} g(x)$ exist, then so does $\lim_{x \rightarrow a} [f(x) + g(x)]$ ” is true or false. Explain your answer.
13. Define $h(2)$ in a way that extends $h(t) = \frac{t^2 + 3t - 10}{(t - 2)}$ to be continuous at $t = 2$.
14. Find $\frac{dy}{dx}$ if $2y = x^2 + \sin y$.
15. Find the absolute extrema of $f(x) = \sqrt{x} + \cos x$ on $\{0, 2\pi\}$.
16. Find the interval on which the function $f(x) = ax^2 + bx + c, a \neq 0$, is increasing and decreasing. Describe the reasoning behind your answer.
17. Find the asymptotes of the graph of $f(x) = -\frac{8}{x^2 - 4}$.
18. Solve $\lim_{x \rightarrow \infty} \frac{2\sqrt{x} + x^{-1}}{3x - 4}$.

Section C

*Answer any **one** question.
The question carries 10 marks.*

19. (a) Suppose $\lim_{x \rightarrow 1} f(x) = 2$ and $\lim_{x \rightarrow 1} g(x) = 9$. Find $\lim_{x \rightarrow 1} \frac{f(x) - x^2 g(x)}{x - f(x)}$.
- (b) Explain the continuity of the function $f(x) = \frac{3x + 4}{x^2 - 4}$.
20. (a) Suppose that $f(-1) = 3$ and $f'(x) = 0$ for all x . Must $f(x) = 3$ for all x ? Give reasons for your answer.
- (b) Discuss the concavity and convexity of the curve $y = x^2$.

(1 × 10 = 10 marks)